

Dushyant Shukla

Software Developer

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Software engineer experienced in real-time interactive simulation systems, performance optimization, and modernizing complex legacy C++ codebases.

Technical Skills

Programming	C++, C, Lua, C#
Graphics Programming	Vulkan, OpenGL, GLSL, RenderDoc, NVIDIA Nsight
Engines & Tools	Proprietary/custom simulation engines, Unity 3D, NVIDIA PhysX SDK, Blender
Software Development	SQLite, AWS, Protocol Buffers, gRPC, Git, CMake, Premake, GitHub Actions, Jira, Confluence

Professional Experience

Mimic Technologies, Inc. (Surgical Science Sweden AB)

Seattle, WA, USA

Software Developer

Jun. 2021 - Present

- Developing VR robotic surgery training scenarios using proprietary simulation engines, Unity3D, and NVIDIA PhysX, enabling surgeons to practice surgical techniques and medical procedures.
- Led cross-studio collaboration to modernize a proprietary legacy C++ medical simulation application, adding support for current OS platforms and enabling renewed product revenue.
- Migrated a proprietary simulation engine from x86 to x64 architecture and implemented a Premake-based build system, improving performance and maintainability.
- Led development of a real-time data pipeline capturing high-resolution surgical performance telemetry (C++, Protocol Buffers, reliable-UDP, SQLite) to support personalized ML-driven feedback.
- Designed and implemented a database migration tool (C++, Dear ImGui) to automate complex schema migrations, replacing an error-prone multi-step manual workflow and ensuring successful contract fulfillment for a key client.
- Contributed to software-hardware integration and validation during product release cycles to ensure system stability and reliability.
- Owned software release workflows, documentation, and code reviews, ensuring reliable product releases and well-maintained technical documentation.

SenecaGlobal IT Services Private Limited

Hyderabad, TG, India

Technical Analyst

Apr. 2019 - Jul. 2019

- Spearheaded the development of a distributed digital asset management workflow using AWS, Spring Boot, and Dropbox APIs, reducing customer processing time and service costs by 50% and enabling a new revenue stream for the client.
- Collaborated with distributed teams and provided off-hours production support, ensuring system reliability and timely issue resolution.

Senior Software Engineer

Sept. 2015 - Mar. 2019

- Led development of data-integration channels for a digital asset management system to syndicate CPG data across Amazon, Walmart, Target, and Kroger using AWS, Spring Boot, Core Java.
- Reduced AWS operating costs by designing low-cost serverless notification and auditing services using AWS Lambda.
- Mentored software engineers, contributing to skill development and team effectiveness.

Education

M.S. in Computer Science, DigiPen Institute of Technology

Apr. 2021

Redmond, WA, USA

B.Tech. in Computer Science & Engineering, Dr. A.P.J Abdul Kalam Technical University

Jun. 2015

India

Projects

Cassini Engine

A custom game-engine (C++, Vulkan, GLSL) | 🏠 Homepage

Jun. 2024 - Present

- Implemented a physically based rendering (PBR) pipeline using the metallic-roughness workflow with glTF asset support.
- Added bindless rendering to reduce draw-call overhead and improve GPU resource access efficiency.
- Developed asynchronous resource loading to prevent stalls and improve runtime performance.